ShockableNet — end-to-end pipeline

From public ECG archives to an alert on a caregiver's phone

Offline — training | **On device**

Data collection

6 PhysioNet databases
CUDB · VFDB — ventricular fibrillation
NSRDB · Fantasia — healthy sinus rhythm
STDB · EDB — cardiac, ambulatory

199 patients · 21,710 windows
split 70/15/15 **by patient**

▶

Preprocessing

resample → 250 Hz
band-pass 0.5–40 Hz, zero phase
normalise by median / IQR
windows 5 s, stride 2.5 s

labelled if one rhythm ≥ 50%
VT split at 150 bpm

▶

ShockableNet

1D CNN · 5 conv blocks + 2 dense
dilation 1·1·2·4·8
mean ‖ max ‖ std pooling
172,498 parameters

focal loss γ=2 · miss penalty 2.5
AdamW · OneCycleLR · 25 epochs

▶

Validation

5-fold cross-validation
folds grouped by patient
test split never read

keep best **specificity**
subject to **sensitivity ≥ 0.90**
(AHA/AAMI floor for VF)

Deployment

float32 · **674 KB**
0.52 ms per window
0.021% of one CPU core

int8 rejected: it silenced one genuinely shockable window

Out-of-fold results · 199 patients · 21,710 windows

ventricular fibrillation	sensitivity	**0.947**	target 0.90	✓
rapid VT (≥ 150 bpm)	sensitivity	**0.845**	target 0.75	✓
normal sinus rhythm	specificity	**0.963**	target 0.99	△
all non-shockable	specificity	**0.963**	target 0.95	✓
discrimination	ROC AUC	**0.982**		✓

Sinus specificity is 0.974 under the AHA evaluation protocol (artifact-free, single-rhythm windows). Both figures are reported together in the model card.

Runtime chain

signal quality gate
▼
ShockableNet
▼
2-of-3 confirmation
▼
alert the caregiver